

# Kunyi (Mark) Ma

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## Education

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**Columbia University**, New York, NY

Ph.D. Student in Mathematics

2024–2029 (to be expected)

M.A. in Mathematics, 4.15/4.

2024–2026

Advisors: Daniela De Silva, Ovidiu Savin

Research interest: PDE Analysis, Free Boundaries

**New York University Shanghai**, Shanghai, China

B.S. in Honors Mathematics and Data Science, magna cum laude, 3.897/4

2020–2024

*Existence of Stationary Measure for Incompressible 2D Stochastic Navier-Stokes Equations* (with Vahagn Nersesyan)

*On modifying Loss Function for PINO to approach Mathematics Nature in Fluid Mechanic* (with Zhuang Su)

**Nanyang Model High School**, Shanghai, China

2017–2020

## Talks

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### Poster Presentations

Workshop on Harmonic Analysis and Partial Differential Equations, University of Pittsburg

Nov. 15th 2025

### Seminar Talks @ Columbia

#### Elliptic PDE Group Meeting

Caffarelli-Cabré: Jensen's Approximation and Uniqueness of viscosity solution to Dirichlet BVP

Feb. 11th 2026

Alt-Caffarelli: Viscosity Solution, Non-degeneracy, Lebesgue Density Estimate

Nov. 7th 2025

Alt-Caffarelli: Euler-Lagrange Equations, Optimal Regularity

Nov. 6th 2025

The Obstacle Problem: Singular Set, Monneau Monotonicity, ACF

Oct. 14th 2025

The Obstacle Problem: Classification of Blowups

Sep. 30th 2025

The Obstacle Problem: Blowups, 2-homogeneity, Convexity

Sep. 23rd 2025

The Obstacle Problem: Minimizers, EL equations, Continuity, Optimal Regularity, Nondegeneracy

Sep. 9th 2025

### The Theory of Bounded Variation Functions & Regularity Analysis of the Minimal Surface PDE

De Giorgi's Lemma for  $C^1$  Caccioppoli Sets

Nov. 6th 2024

De Giorgi's Lemma for Harmonic and  $C^1$  function

Oct. 16th 2024

Regularity of the reduced boundary

Sep. 25th 2024

Basic properties of functions of bounded variation

Sep. 11th 2024

### Informal Reading Seminar on GR

Local Theory for Quasi-linear Wave Equations

Nov. 18th 2024

Existence of Solutions to non-constant coefficient linear wave IVP

Oct. 7th 2024

Representation formula using Homogeneous Distribution

Sep. 16th 2024

## Teachings

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### Teaching Assistant

Applied Functional Analysis APMA4150, Columbia University

Spring 2026

Ordinary Differential Equations UN2030, Columbia University

Spring 2026

Analysis & Probability I GR6151 (Graduate Level), Columbia University

Fall 2025

Linear Algebra S2010, Columbia University

Summer 2025 B

Analysis and Optimization S2500 (Grader), Columbia University

Summer 2025 B

Introduction to Modern Analysis I S4061, Columbia University

Summer 2025 A

### Directed Reading Program

Introduction to Elliptic PDEs and Geometric Measure Theory (with Liam Dennehy, Marlet Ma, Barnett Pesin) Spring 2026

Student Tutor

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|---------------------------------------------------------|-----------|
| College Algebra and Calculus I–III, Columbia University | Fall 2024 |
| ARC Learning Assistant for Pre-Calculus, NYU Shanghai   | Fall 2021 |

Honors and Awards

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NYU Shanghai

Magna cum laude; Phi Beta Kappa; NYU Shanghai Excellence Award; NYU Honors Scholar: Founders Day Award; 2023 NYU Shanghai Dean’s Undergraduate Research Funding; 2022 NYU Shanghai Hack the Pearl, Champion of Data Modelling Track; 2020–2022 NYU Shanghai Recognition Award; 2020–2023 NYU Shanghai Dean’s List

Attendance

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|-------------------------------------------------------------------------------------------|----------------|
| SIAM Conference on Analysis of Partial Differential Equations (PD25)                      | Nov. 2025      |
| Workshop on Harmonic Analysis and Partial Differential Equations, University of Pittsburg | Nov. 2025      |
| The Third Joint SIAM/CAIMS Annual Meetings (AN25), Montréal, Québec, Canada               | July 2025      |
| PDE Summer School 2025 Theory and Applications of Elliptic PDE, UC Irvine                 | June 2025      |
| Rocky Mountain Mathematics Consortium Summer School 2025, ASU, University of Wyoming      | June 2025      |
| Duke Mathematics Journal 2025 Conference, Duke University                                 | Apr. 2025      |
| Winter School: Boundary and Singularity in Fluid Mechanics, Simons Center                 | Jan. 2025      |
| Differential Geometry Summer School 2024, PKU BICMR                                       | July-Aug. 2024 |
| Workshop on Minimal Surfaces and Mean Curvature Flow 2024, PKU BICMR                      | July 2024      |
| International Congress of Chinese Mathematicians 2022, THU YMSC                           | July-Aug. 2022 |

Extracurricular

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|--------------------------------------------------------|---------------------|
| Orientation Ambassador, NYU Shanghai                   | May–Sep. 2021       |
| President, Literature@NYUSH Student Club, NYU Shanghai | Jun. 2021–May 2022  |
| Tenor, NYU Shanghai Elite A Cappella Chamber Singers   | Feb. 2021–Dec. 2022 |
| Quantitative Researcher, Ipsos China, Shanghai         | Jun.–Aug. 2021      |